Orbital Dragons to Hunt Cryptids: Utilizing Non-Euclidean Geometry to Solve Busy Beaver

Enan Srivastava

DeepMath 2026 — Extended Abstract

Abstract

The fifth Busy Beaver number was settled in 2024; the sixth is guarded by a residual set of "cryptid" Turing machines whose halting problems resist every known decider. We present a certified pipeline that hunts these cryptids geometrically: raw tape dynamics are compressed into *orbit events*, embedded in non-Euclidean (logarithmic, simplex, hyperbolic) charts where their self-similar structure becomes affine, and then re-certified in core Lean 4 (axioms $\{\text{propext}, \text{Quot.sound}\}$ only). The headline result: two previously open BB(6) machines are reduced, machine-checked end to end, to conditional theorems $\mathsf{SafeMetaOrbit} \to \neg\mathsf{RawHalts}$, where $\mathsf{SafeMetaOrbit}$ concerns a deterministic affine map on five integers. Every raw step either machine ever takes—startup, one degenerate episode, and an eternity of seven certified episode types—is accounted for by theorems; the remaining conjecture is a subtractive Euclid/Collatz hybrid whose excluded parameter bands are *provably lethal*, so avoiding them is exactly the halting question. The proofs were compiled by generator–verifier loops: fail-closed empirical calculi validated against 10^9 raw steps, then emitted as Lean and re-checked by the kernel.

1 Cryptids and the shape of the residual set

After BB(5) = 47,176,870 was machine-checked, attention moved to the BB(6) frontier, where a residual cohort of holdout machines—"cryptids" such as Antihydra—encode Collatz-like arithmetic. We classified 1,094 holdouts with a Collatz-lens profiler (growth dimension, record structure, exact projections); none is decided by existing budgets, and 51 exhibit *compact growth*: tape configurations that stay describable by a bounded vector of counters. These are the machines our pipeline targets.

2 Incision geometry: checking only at the boundary

For a compact-growth machine, halting danger concentrates at *incision events*: the rare configurations where a distinguished "probe" state reads the symbol that could launch the halting transition. In Lean we prove this is not a heuristic but an equivalence: for Antihydra, an event process eorbit (a closed-form *volley* skipping through the interior dynamics) satisfies

$$\mathsf{SafeOrbit} \iff \forall k,\ \mathsf{eorbit}(k) \neq \bot,$$

so checking safety *only at incisions* is sound and complete. Embedding event clouds via $x \mapsto \log(1+x)$, projective simplex normalization, and upper-half-plane charts renders the record dynamics affine and exposes family structure across the cohort—the empirical atlas that guides which machines admit exact certification.

3 Orbital dragons: predictive emulation

An *orbit finder* fits fail-closed exact rational affine specifications to record-level dynamics (exact arithmetic, no tolerance; any residual rejects the spec). On the nine-machine certification queue it returns exact specs for four machines: Antihydra, one known reduction target, and two machines new to this program ("dossier-2/3," a mirror pair). A generic predictor then replays orbits at record granularity, giving certified skip horizons and, crucially, the complete grammar of each machine's motion.

4 The interaction calculus and conditional closure

For the mirror pair we decode the entire eternal dynamics into a *local interaction calculus* on a tape shape $1^a\,0\,1^b\,0\,1^c$: a ten-step "caterpillar" gait crawls the field; static holes are *crossed* (+4) or *terminated* (+6) according to arrival parity; adjacent holes *fuse* (+9); the field edge runs a parity-dependent business. Validated episode-for-episode against 10^9 raw steps per machine, the calculus compiles to seven Lean episode theorems per machine over a uniform `holeTape` shape, glued by an n-fold cycle induction and a mixed sweep lemma whose F-parity constraint explains every +4/+6/+9 shift. A five-integer abstract orbit metaStep dispatches the episodes; transport, a segment keystone, and fully concrete startup anchors (235 and 2,044 certified steps, including one degenerate episode each) yield, at axioms {propext, Quot.sound}:

Theorem 1. *For each of the two machines,* $\mathsf{SafeMetaOrbit} \longrightarrow \neg\,\mathsf{RawHalts}(\mathsf{blankStart})$.

SafeMetaOrbit states that the abstract orbit never leaves its guarded region. The renewal map is a subtractive Euclid/Collatz hybrid: generic loops conserve $\gamma = c - b$ while a grows by 6; deep renewals swap (a, b); fusions rotate $(a, 2, c) \mapsto (a{+}9,\ c{-}6,\ 1)$. Simulating the excluded bands ($\gamma \in \{1, 3, 5\}$ at renewal, $b \le 2$ at carried entries) produces a halt within $\sim$50 steps: the conjecture *is* the halting problem, compressed from a Turing machine to arithmetic on five integers. Both orbits survive 10^8 abstract episodes ($\approx 10^{15}$ raw steps).

5 Method: generator–verifier proof compilation

All certified layers are emitted by simulation-driven compilers: atoms are derived by running the machine table over declared precondition windows (the theorem cannot diverge from the machine); episodes are compiled from segment plans with funext normalizations between stages; mirror machines are generated by systematic sign-flips of the plans. The empirical layer is fail-closed—any unexplained episode halts the pipeline—and everything emitted is re-checked by the Lean kernel, so the generators are convenience, not trust surface. Notable proof-engineering findings: `omega` leaks Classical.choice from negated-conjunction hypotheses; and rewriting on `stepN` goals triggers a post-rewrite definitional-equality attempt that evaluates entire runs (a hidden quadratic cost), eliminated by term-mode run gluing.

6 Honest scope

No unconditional non-halting claim is made for any open machine. Four machines (Antihydra, its companion, and the mirror pair) are reduced to single conjectures; for the mirror pair the conjecture is a deterministic five-integer recurrence, sharper in kind than its Collatz-like relatives. Empirical horizons are bounded by computation, not by what is true.

Acknowledgments. The Lean developments, empirical calculi, and proof compilers were built in close collaboration with an AI assistant (Claude, Anthropic); all emitted proofs are machine-checked by the Lean 4 kernel with the stated axiom bar, and all empirical claims were validated by differential testing against independent raw-machine emulation.

References

[1] The bbchallenge Collaboration. *Determination of BB(5)*, machine-checked in Rocq (Coq-BB5), 2024.

[2] The bbchallenge project. `https://bbchallenge.org` — cryptids and the BB(6) frontier.

[3] E. Srivastava. *bbchallenge-proofs*: certified emulation pipeline (busy-beaver-core, dossier, flying-dragon packages), 2026. `https://github.com/Enan456/bbchallenge-proofs`.